KrishiVed Market Decision Support — Basic SRS & Version 1 Implementation Plan

A. Document Control and Purpose

Attribute	Value
Project name	KrishiVed Market Decision Support
Competition	AI Synergy Hackathon 2026 — Round 1
Selected case	Case 02 — Agriculture: The Harvest Decision Window
Document title	Basic Software Requirements Specification and Version 1 Implementation Plan
Version	1.0
Status	Proposed / Planned
Date	2026-10-14
Purpose	Define the software requirements, architecture, acceptance criteria, and implementation plan for a Version 1 transport-adjusted mandi comparison tool, suitable for evaluation by competition judges and for use as a build specification in Round 2.
Intended audience	Competition evaluators; Round 2 development team
Relationship to Round 1 submission	This document elaborates the SRS, workflow, AI/human responsibility model, acceptance tests, and implementation plan already presented in `SUBMISSION.md`. It introduces no new features, datasets, integrations, or test results beyond what the Round 1 submission documents establish. Where a detail is not fixed by existing project files, it is labeled as a proposal.

Integrity statement: No team-member names, approvals, field trials, API integrations, benchmark results, or successful test outcomes are claimed in this document. All six acceptance tests carry status "Not run."

B. Users and Stakeholders

Stakeholder	Category	Role	Information needs	Interaction with the system
Smallholder vegetable farmer	Primary user	Decides when and where to sell perishable produce	Net proceeds per mandi after selling costs; quote freshness; weather context; assumptions affecting estimates	Enters crop, quantity, location, candidate mandis, cost estimates; reviews comparison and warnings; makes final selling decision
Local aggregator / transporter	Supporting stakeholder	Transports produce; validates cost estimates	Accurate transport and loading cost information for a given route and volume	May assist the farmer with data entry; validates transport-cost inputs; may view the same comparison output
Mandi trader	Supporting stakeholder (data source)	Sets or references market prices	Not a system user	System may source quotes attributed to traders (simulated in V1); trader does not interact with the system

Stakeholder	Category	Role	Information needs	Interaction with the system
Agriculture extension worker	Supporting stakeholder	Helps interpret weather and agronomic context	Weather forecast source, coverage, and limitations; crop-quality context	May help the farmer interpret weather warnings; not granted direct system access in V1
Farmer Producer Organisation (FPO)	Supporting stakeholder	Collective decision-making, group selling	Aggregated comparison across members (future)	No V1 interaction; group-comparison feature is explicitly out of scope
System administrator / data maintainer	Operator	Configures data sources and thresholds	Data source status; freshness-threshold configuration; fixture integrity	Configures freshness threshold (default 4 hours), manages simulated fixture data; no access to farmer inputs

Access model: No login or account creation is required for the primary farmer user. The V1 system collects no personally identifiable information (PII).

C. System Scope and Boundaries

C.1 Problem Being Solved

Smallholder vegetable farmers lack a reliable, accessible way to compare the expected net proceeds of selling a perishable crop across nearby mandis when quoted prices, transport costs, weather conditions, and connectivity vary. A mandi offering the highest quoted price may not produce the highest net proceeds after selling costs.

C.2 Intended Purpose

Provide a transparent, deterministic comparison of estimated net proceeds across a small set of candidate mandis, with visible data-quality warnings and understandable explanations, while preserving farmer decision authority.

C.3 In-Scope Features (Version 1)

#	Feature
1	Accept crop, quantity, origin location, and candidate mandis (up to 5)
2	Accept or load quoted prices, source metadata, and timestamps
3	Accept transport, loading, and unloading cost estimates
4	Calculate selling cost per kg, net realization per kg, and total expected net proceeds deterministically
5	Compare net proceeds across candidate mandis where data is sufficient
6	Display quote source, timestamp, and freshness status for every price
7	Flag missing, stale, conflicting, or unavailable data explicitly
8	Show weather/context information with source, update time, and geographic coverage label when available
9	Generate plain-language AI-assisted explanations of validated comparison results
10	Display offline/cached data with explicit freshness warnings
11	English-language interface

C.4 Out-of-Scope Features (Version 1)

- Price prediction or forecasting
- Harvest-timing optimization or crop-quality degradation modelling
- Autonomous selling, purchasing, or logistics booking
- Collective selling or group transport coordination
- SMS, IVR, WhatsApp, or voice interfaces
- Hindi or other language interfaces
- Nationwide mandi or crop coverage
- Guaranteed selling outcomes (price, income, yield, or waste reduction)
- Account creation, PII collection, or server-side storage of farmer inputs

C.5 External Systems and Dependencies

Dependency	Status	Classification
Simulated market fixture data (`tests/fixtures/*.json`)	Exists in repository	**Confirmed existing artifact**
Deterministic calculation engine	Specified; not yet implemented	**Proposed design**
Mandi price API (eNAM, state APMC)	Not connected; access not verified	**Future integration requiring verification**
Weather API (IMD, OpenWeatherMap)	Not connected; data structure defined	**Future integration requiring verification**
AI explanation service (LLM)	Prompt designed; not deployed	**Proposed design**
Offline cache (LocalStorage / Service Worker)	Behavior specified in acceptance tests; not implemented	**Proposed design**

C.6 Version 1 Boundaries

Boundary	Value
Crop coverage	One crop per session
Mandi coverage	Up to 5 candidate mandis
Language	English only
Data source (development/testing)	Simulated fixtures, clearly labeled
Data source (prototype, if verified)	Live mandi API — not confirmed as integrated
Connectivity	Offline-capable with cache; no guarantee of complete offline operation
Decision authority	Farmer decides; system never recommends a mandi

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D. Inputs and Outputs

D.1 System Inputs

Field / Input	Description	Data type / format	Required	Validation rules	Failure behaviour if missing or invalid
`crop`	Crop being sold	String (enum: tomato, etc.)	Yes	Must match a supported crop identifier	Reject submission with message: "Select a crop."
`quantity_kg`	Quantity available to sell	Number (kg)	Yes	Must be > 0; unit explicitly kg	Reject if ≤ 0 or missing: "Quantity must be greater than zero."

Field / Input	Description	Data type / format	Required	Validation rules	Failure behaviour if missing or invalid
`origin_location`	Farmer's village or cluster	String	Yes	Village/cluster level only; no GPS, no PII	Reject if empty: "Enter your village or cluster."
`candidate_markets`	List of mandis to compare (1–5)	Array of market objects	Yes	Array length 1–5	Reject if empty or > 5: "Select between 1 and 5 mandis."
`market_id`	Unique market identifier	String	Yes	Unique within submission	Reject duplicates.
`market_name`	Display name of mandi	String	Yes	Non-empty	Reject if empty.
`distance_km`	Distance from origin to mandi	Number (km)	Proposed field — not finalized in source material	Must be ≥ 0 if provided	If missing, proceed; distance not used in core calculation (costs are entered directly).
`quoted_price_per_kg`	Quoted price from source	Number (₹/kg)	Yes (for comparison)	Must be > 0; unit must be ₹/kg	If null: mark market `data_freshness: "missing"`; exclude from ranking; display "—". Never invent a value.
`quote_source`	Origin of the quote (e.g., "simulated", "trader", "eNAM")	String	Yes	Non-empty; provenance must be preserved	If missing: flag as "source unknown"; do not present as verified.
`quote_timestamp`	ISO 8601 timestamp of quote	ISO 8601 string	Yes (for freshness check)	Valid ISO 8601; parsed against current time	If null: mark freshness "unknown"; display warning "Quote age cannot be determined."
`transport_cost_total`	Total transport cost for the trip	Number (₹ total)	Yes (for ranking)	≥ 0; user-entered estimate or cached value	If missing: prompt user. If user declines: mark market "costs unknown"; exclude from net-proceeds ranking. Never default to zero.
`loading_cost_total`	Loading charges	Number (₹ total)	No	≥ 0 if provided; default 0 only if explicitly confirmed unknown-and-zero by user	If not provided and user did not confirm zero: treat as unknown; exclude from ranking.
`unloading_cost_total`	Unloading charges	Number (₹ total)	No	≥ 0 if provided	Same as loading cost.
`freshness_threshold_hours`	Configurable max quote age	Number (hours)	Yes (config)	> 0; proposed default 4 hours	System default applied if not set. Policy is a proposal, not a universal standard.

Field / Input	Description	Data type / format	Required	Validation rules	Failure behaviour if missing or invalid
`weather_forecast`	Weather context object	JSON object	No	If present: must include `condition`, `source`, `source_timestamp`, `geographic_coverage`	If null or service fails: display "Weather data unavailable" with last-attempt timestamp; proceed without weather recommendation.
`connectivity`	Current connectivity state	Enum: "online" / "offline"	No	—	If offline: use cache if valid; display offline banner; allow manual entry.

D.2 System Outputs

Output	Description	Unit / format	Source / calculation method	Associated warnings or limitations
`selling_costs_per_kg`	Total included selling costs divided by quantity	₹ / kg	`(transport + loading + unloading) ÷ quantity_kg`	"Based on entered estimates; actual costs may differ."
`net_realization_per_kg`	Quoted price minus selling cost per kg	₹ / kg	`quoted_price_per_kg – selling_costs_per_kg`	"Estimate only; does not account for cultivation cost, losses, or unmodelled charges."
`total_net_proceeds`	Quantity multiplied by net realization per kg	₹ total	`quantity_kg × net_realization_per_kg`	"Estimated proceeds after included selling costs — not guaranteed."
`data_freshness`	Freshness status of each quote	Enum: `current` / `stale` / `missing` / `unknown`	Compared against `freshness_threshold_hours`	Stale: "Quote is N hours old (threshold: Xh)." Missing: "No quote available."
`ranked_comparison`	Markets ordered by total net proceeds (where sufficient data)	Ordered list	Deterministic sort on `total_net_proceeds`	Markets with missing quotes or unknown costs are excluded from ranking, shown separately.
`warnings`	Per-market data-quality warnings	Array of strings	Rule-based checks	Warnings never suppressed.
`assumptions_panel`	Collapsible list of assumptions affecting the comparison	Text list	System-generated from input metadata	Includes: transport estimate provenance, quote freshness, weather coverage, simulation labeling.
`ai_explanation`	Plain-language explanation of validated results	English paragraph (≤200 words proposed)	AI layer receiving only validated deterministic output	Must preserve all numbers, units, and caveats from engine output. Must not recommend a mandi.

Output	Description	Unit / format	Source / calculation method	Associated warnings or limitations
`weather_context`	Weather condition, probability, source, timestamp, coverage	JSON + display text	Weather source (simulated or future API)	Must show geographic coverage label; must not be used to automate decisions.
`offline_banner`	Connectivity status indicator	Text + timestamp	Cache metadata	"Offline — showing cached data from [timestamp]." Never claims live data when cached.

E. Functional Requirements

Preserves the intent of existing FR-01 through FR-10 from `specs/01-solution-srs.md` and `SUBMISSION.md`.

ID	Requirement statement	Rationale	Inputs	Expected output	Priority	Verification method	Dependencies
FR-01	The system shall accept crop, quantity, origin location, and selling-window inputs from the farmer.	Core data entry enabling all downstream calculation.	`crop`, `quantity_kg`, `origin_location`, selling window	Validated input record	Must	Valid and invalid input tests (boundary values: 0, negative, missing fields)	None
FR-02	The system shall display mandi quotes with their source and timestamp for every candidate market.	Transparency requirement (E01: quote obtainability uncertain; E04: trust gap).	`quoted_price_per_kg`, `quote_source`, `quote_timestamp`	Quote display with provenance metadata and freshness status	Must	Inspect output against source records; AT-03, AT-04	FR-01
FR-03	The system shall calculate estimated net realization per kg after all included selling costs, using deterministic arithmetic.	Correctness requirement (NFR-01). LLM arithmetic is unreliable.	`quoted_price_per_kg`, `transport_cost_total`, `loading_cost_total`, `unloading_cost_total`, `quantity_kg`	`selling_costs_per_kg`, `net_realization_per_kg`	Must	AT-01: verify ₹19/kg for specified inputs	FR-01, input cost fields
FR-04	The system shall compare estimated total net proceeds across candidate mandis where sufficient data exists.	Core value proposition: transport-adjusted comparison (E01 + E02).	Outputs of FR-03 for each market	Ranked comparison list; markets with insufficient data excluded and flagged	Must	AT-01, AT-02: verify ranking matches net-proceeds order, not quoted-price order	FR-03

ID	Requirement statement	Rationale	Inputs	Expected output	Priority	Verification method	Dependencies
FR-05	The system shall identify missing, stale, conflicting, or unavailable data and display explicit warnings.	No-fabrication rule; case boundary prohibits invented data.	`quote_timestamp`, `freshness_threshold_hours`, null checks, conflict detection	`data_freshness` status per market; `warnings` array	Must	AT-03 (missing), AT-04 (stale), error-handling tests for conflict	FR-02
FR-06	The system shall explain the comparison and its key assumptions in plain language via the AI explanation layer.	Trust requirement (E04): explanations must be transparent and preserve farmer authority.	Validated comparison JSON (deterministic output only)	English explanation paragraph preserving all numbers and caveats	Must	AT-06 (explanation consistency check: explanation numbers == engine numbers)	FR-04, AI service or template fallback
FR-07	The system shall show relevant weather information with its source, update time, and geographic coverage label when available.	Weather context (E05) relevant but uncertain (U05).	`weather_forecast` object or null	Weather display with source/timestamp/coverage; or "unavailable" message	Should	Manual inspection; AT-06 (weather missing case)	Weather data source (simulated or API)
FR-08	The system shall provide an English-language interface.	V1 scope decision (A04); Hindi deferred.	UI text resources	English interface	Should	Language inspection	None
FR-09	The system shall display cached information with a clear freshness warning and never present it as live data.	Connectivity requirement (E03: 2 of 5 villages unreliable).	Cache metadata (`quote_timestamp`, cache-time)	Offline/cached banner with timestamp; calculations still functional	Should	AT-05: offline with cached data	Cache layer (LocalStorage or equivalent)
FR-10	The system shall provide selling-time comparisons using crop-condition and storage information.	Not achievable in V1 (U04: quality loss data unknown).	Crop condition, storage status, delay duration	Sell-now vs delay comparison	Future	Not applicable in V1	Requires crop-quality research data — not available

F. Non-Functional Requirements

ID	Requirement	Category	Target / threshold	Classification	Verification method
NFR-01	The deterministic calculation engine shall return reproducible results for identical valid inputs.	Correctness	100% match on AT-01, AT-02 manual verification	Confirmed requirement	Manual recalculation; unit tests
NFR-02	Every displayed estimate shall identify its data source, timestamp, included costs, and material assumptions.	Transparency	All outputs auditable against input metadata	Confirmed requirement	Output inspection against input records
NFR-03	A farmer shall be able to understand the comparison without interpreting technical terminology.	Usability	Proxy-user test: >80% success identifying higher-net market and key assumption	**Proposed target** — requires validation	Evaluation plan: proxy user walkthrough of AT-02 scenario
NFR-04	The system shall handle failed data requests, connectivity loss, and unavailable quotes without fabricating replacements.	Resilience	AT-03, AT-05 pass; zero fabricated values in any output	Confirmed requirement	AT-03, AT-05 execution
NFR-05	The system shall minimize personal data collection and avoid unnecessary tracking.	Security / privacy	No login, no PII, no server-side storage of farmer inputs	Confirmed requirement	Code inspection; data-flow review
NFR-06	The interface shall be usable on mobile screens with limited technical confidence and low-end devices.	Accessibility	Functional at 320px viewport width; compatible with 2G-level bandwidth simulation	**Proposed target** — requires validation	Device testing (Build Round)
NFR-07	Market retrieval, calculation, and explanation layers shall be separable for independent modification and testing.	Maintainability	Module boundaries enforceable; calculation engine testable with zero external dependencies	Confirmed requirement	Architecture review; unit test isolation
NFR-08	Translated explanations (future) shall preserve numerical values, units, caveats, and uncertainty.	Language reliability	Round-trip numerical accuracy: 100% for future Hindi output	**Proposed target** — future scope	Future: bilingual round-trip comparison test

ID	Requirement	Category	Target / threshold	Classification	Verification method
NFR-09	The system shall render comparison results within a proposed target of 2 seconds on a mid-range mobile device for 5 markets.	Performance	2 seconds (proposed)	**Proposed target — not measured; requires validation**	Performance test in Build Round

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G. End-to-End Workflow

G.1 Normal Flow

1. Farmer opens the interface (English, mobile).
2. Farmer enters crop, quantity, origin location, and selects candidate mandis (1–5).
3. For each mandi, the system loads or the farmer enters: quoted price, quote source, quote timestamp, transport cost, loading cost, unloading cost.
4. System validates inputs (FR-01): checks for missing fields, invalid quantities (≤ 0), unit mismatches.
5. System performs data-quality checks (FR-05): evaluates freshness against threshold (proposed: 4 hours), detects missing quotes, conflicting values, unavailable costs.
6. For each market with sufficient data, deterministic engine calculates (FR-03):
   - Selling cost per kg = Total applicable selling costs ÷ Quantity in kg
   - Net realization per kg = Quoted price per kg − Selling cost per kg
   - Total expected net proceeds = Quantity in kg × Net realization per kg
7. System ranks markets by total net proceeds where data is sufficient (FR-04).
8. AI explanation layer receives validated comparison JSON and generates plain-language explanation (FR-06).
9. Weather context is displayed if available, with source, timestamp, and coverage label (FR-07).
10. Interface presents: comparison table, warnings, assumptions panel, explanation, weather context, data-freshness status.
11. Farmer reviews all information, verifies practical details (actual quote obtainability, crop condition), and makes the final selling decision. The system does not recommend a mandi.

G.2 Worked Example (AT-01 Inputs)

Market	Quoted price	Transport	Loading	Unloading	Quantity	Selling cost/kg	Net realization/kg	Total net proceeds
Mandi A	₹24/kg	₹300	₹100	₹100	100 kg	₹5.00	**₹19.00**	**₹1,900**

These are example inputs for verification purposes, not market recommendations or claims about actual prices.

G.3 AT-02 Verification (Transport Erases Quoted-Price Advantage)

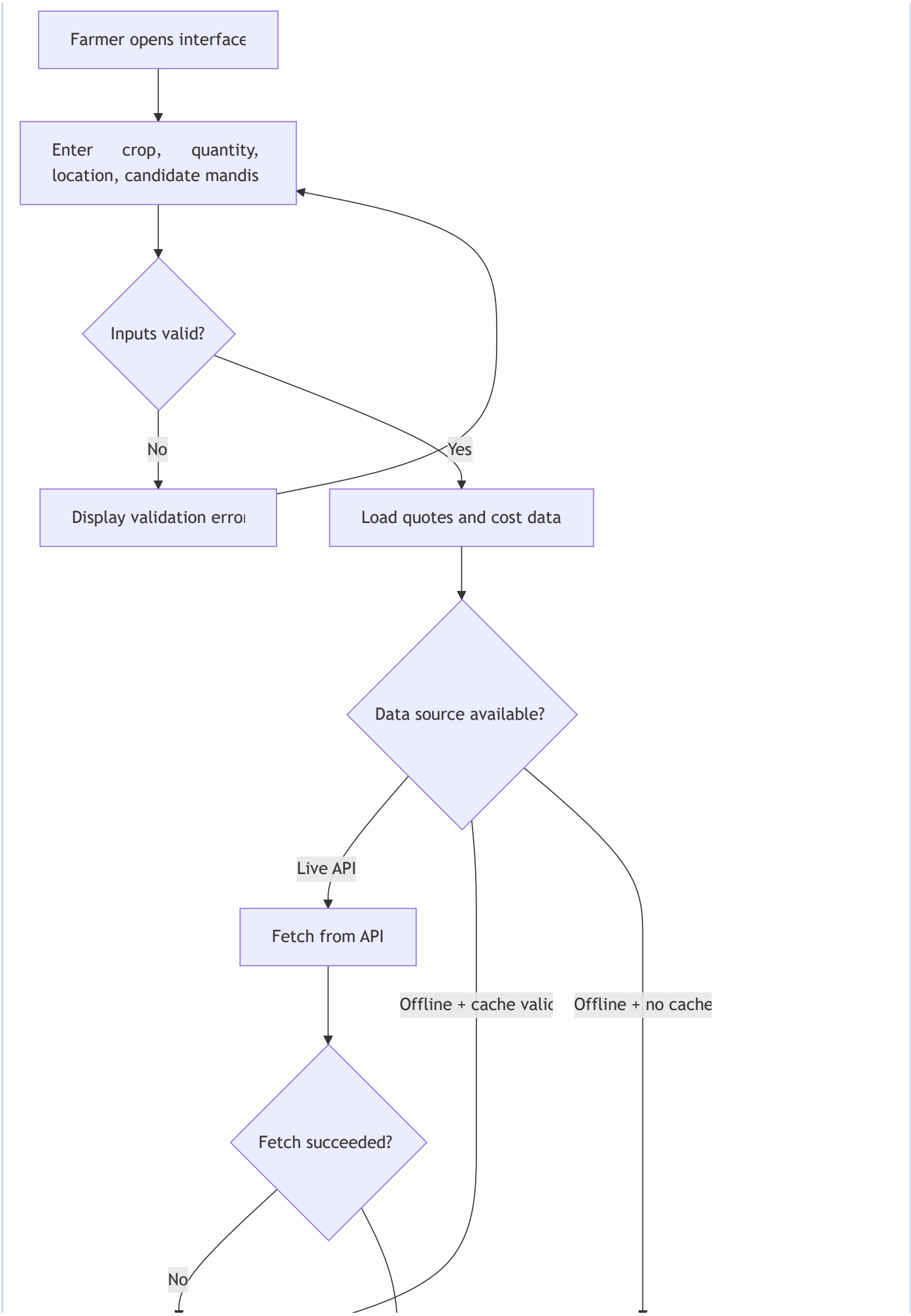
Market	Quoted price	Total costs	Selling cost/kg	Net realization/kg	Total net proceeds	Rank
Mandi A	₹22/kg	₹200 (100+50+50)	₹2.00	**₹20.00**	₹2,000	**1**
Mandi B	₹24/kg	₹500 (300+100+100)	₹5.00	**₹19.00**	₹1,900	2

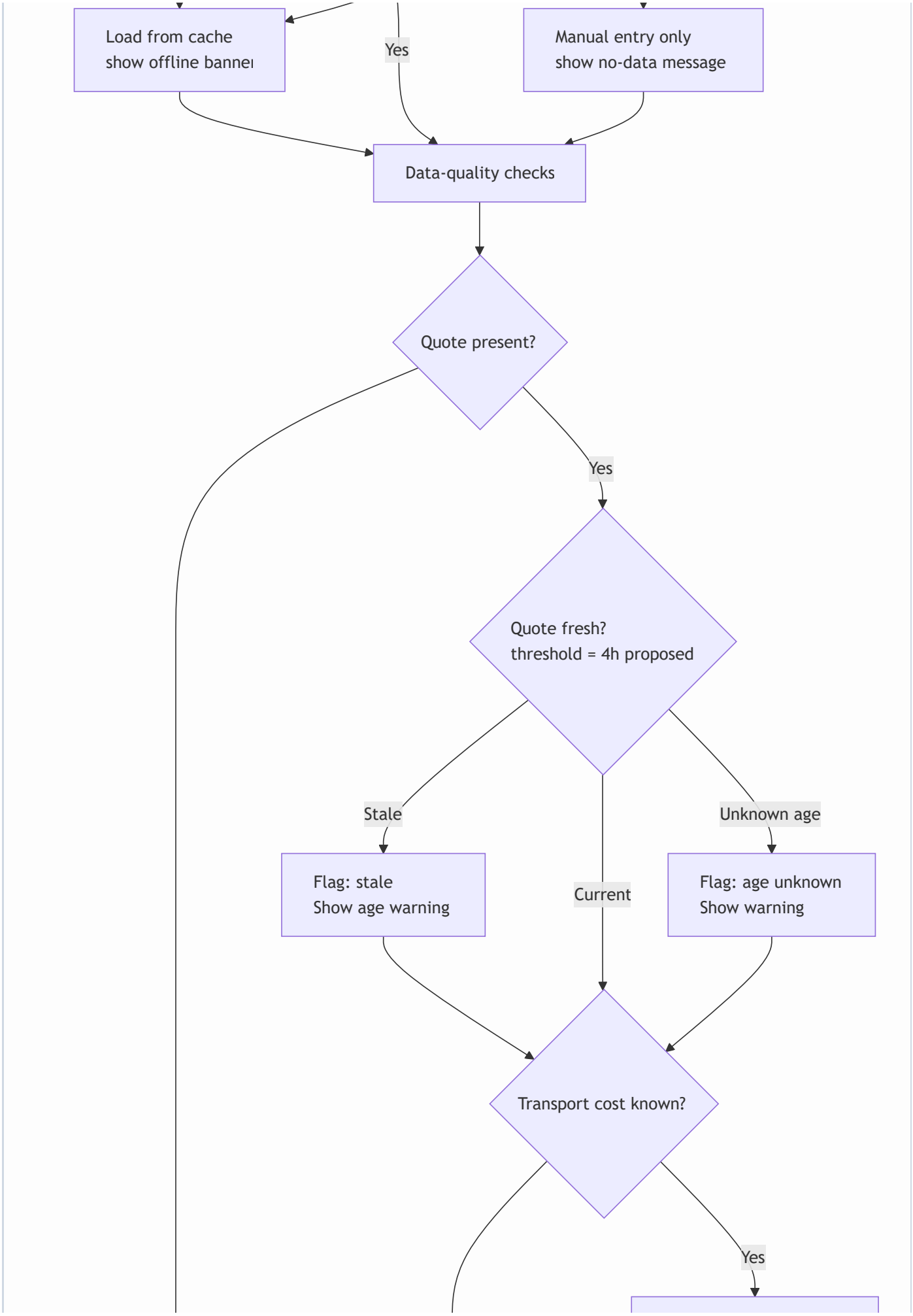
Result: Mandi A wins on net proceeds despite Mandi B's higher quoted price. The system must not rank Mandi B as better based on quoted price alone.

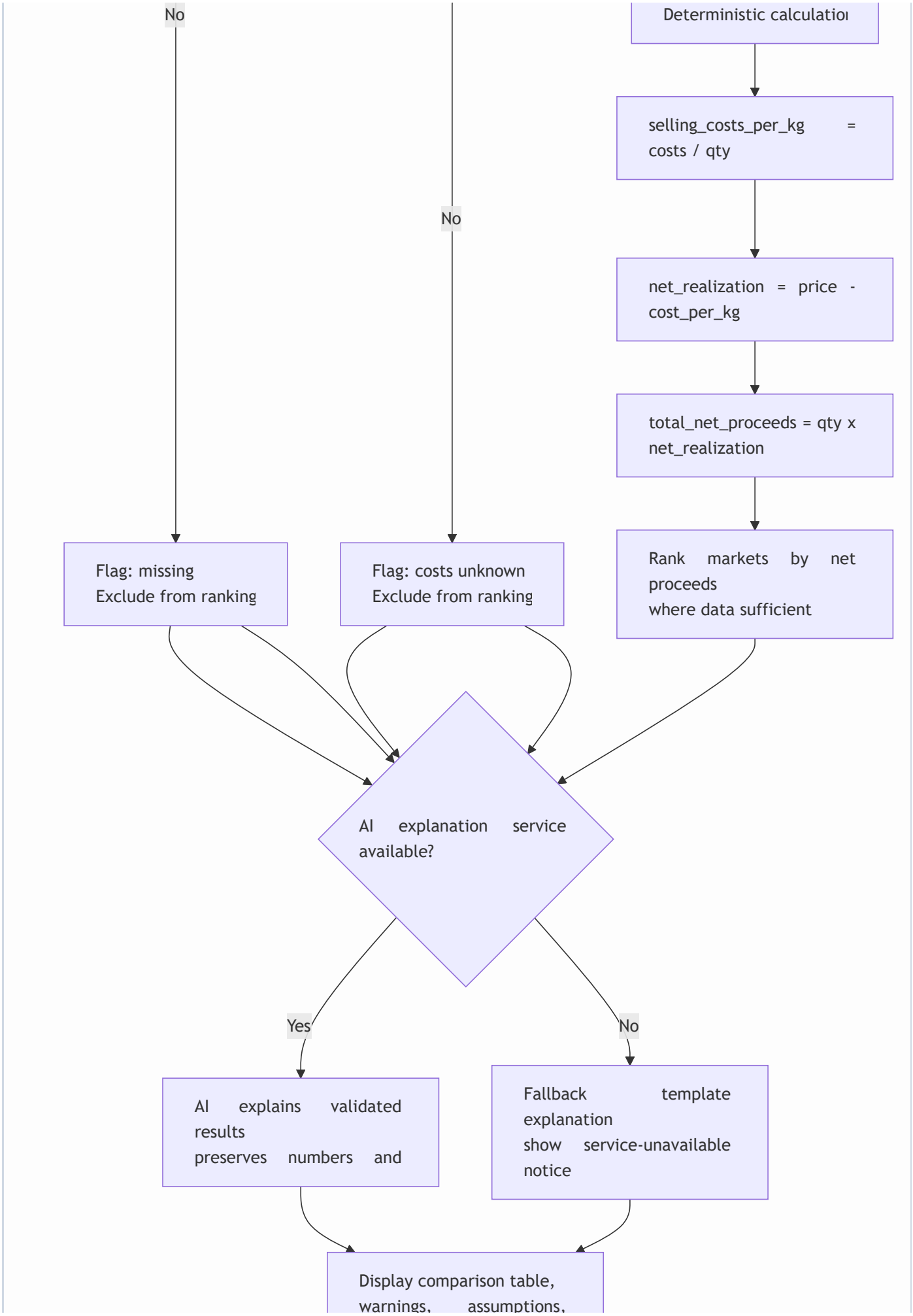
G.4 Failure Flows

Flow	Trigger	System path
Missing-data flow	`quoted_price_per_kg` is null for one market	Step 4 passes → Step 5 flags `data_freshness: "missing"` → Step 6 skips this market → Step 7 excludes from ranking → Step 10 shows "—" and warning → Step 11 continues with remaining markets
Stale or conflicting quote flow	`quote_timestamp` exceeds freshness threshold, or two sources report different prices for same market	Step 5 flags stale ("Quote is N hours old") or conflict ("Two sources disagree") → Step 6 may still calculate but marks warning → Step 7 ranks only current non-conflicting data → Step 10 displays warning prominently → No auto-resolution
Unknown-cost flow	`transport_cost_total` missing and user declines to enter	Step 4 accepts partial data → Step 5 marks "costs unknown" → Step 6 cannot calculate net proceeds for this market → Step 7 excludes from ranking → Step 10 shows "Cannot compare: selling costs unknown"
Network / external-service failure flow	Live API fetch fails; no cache available	Step 3 falls back to manual entry → banner "No live data available; enter quotes manually" → workflow continues from Step 4 with user-entered data
AI-service failure flow	Explanation service unavailable or times out	Step 8 fails → fallback template activated: "Market A nets ₹X/kg, Market B nets ₹Y/kg. Key assumption: transport costs as entered." → deterministic table from Step 7 still displayed → Step 10 shows fallback message + "Service unavailable — review numbers directly"
Cached-data flow	Offline; valid cache exists (age < threshold)	Step 3 loads from cache → Step 5 marks source "cached" with cache timestamp → banner "Offline — showing cached data from [timestamp]" → Steps 6–7 proceed → Step 10 must not present cached data as live

G.5 Workflow Diagram (Mermaid)


Farmer opens interface
Enter crop, quantity, location, candidate mandis
Inputs valid?
No
Yes
Display validation erro
Load quotes and cost data
Data source available?
Live API
Fetch from API
Offline + cache valic
Offline + no cache
Fetch succeeded?
No


Load from cache
show offline banner
Yes
Manual entry only
show no-data message
Data-quality checks
Quote present?
Yes
Quote fresh?
threshold = 4h proposed
Stale
Flag: stale
Show age warning
Current
Unknown age
Flag: age unknown
Show warning
Transport cost known?
Yes


No
No
Deterministic calculatio
selling_costs_per_kg = costs / qty
net_realization = price - cost_per_kg
total_net_proceeds = qty x net_realization
Rank markets by net proceeds where data sufficient
Flag: missing Exclude from ranking
Flag: costs unknown Exclude from ranking
AI explanation service available?
Yes
No
AI explains validated results preserves numbers and
Fallback template explanation show service-unavailable notice
Display comparison table, warnings, assumptions,

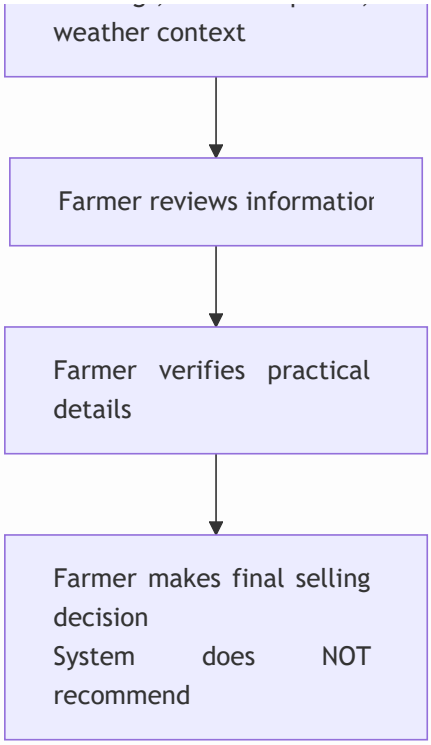


Diagram note: This diagram represents the proposed V1 workflow. No branch in this diagram implies implemented functionality; all paths describe intended behavior for Build Round implementation.

H. AI Responsibilities and Human Responsibilities

Task	Deterministic software responsibility	AI responsibility	Human (farmer) responsibility	Safety / validation rule
Accept and validate inputs	Parse, type-check, range-check all fields; reject invalid inputs	None	Provide accurate inputs	Invalid inputs rejected before any calculation
Retrieve or load market quotes	Fetch from source; preserve source, timestamp, provenance	None	Enter manually if no source available	Missing quotes flagged, never invented
Assess data freshness	Compare `quote_timestamp` against threshold (proposed 4h); compute age	None	Note warning displayed	Stale data flagged with visible age
Detect missing / conflicting data	Null checks; cross-source comparison; flag conflicts	None	Acknowledge warnings	Conflicts surfaced, never auto-resolved
Calculate selling costs per kg	`(transport + loading + unloading) ÷ quantity`	None	Provide or confirm cost estimates	Missing costs never treated as zero silently
Calculate net realization per kg	`quoted_price − selling_costs_per_kg`	None	Review result	Pure arithmetic; zero LLM involvement
Calculate total net proceeds	`quantity × net_realization_per_kg`	None	Review result	Pure arithmetic; zero LLM involvement

Task	Deterministic software responsibility	AI responsibility	Human (farmer) responsibility	Safety / validation rule
Rank markets	Sort by `total_net_proceeds` where data sufficient	None	Review ranking	Markets with missing data excluded, not ranked by default
Explain comparison results	—	Generate plain-language explanation from validated comparison JSON only	Read and interpret	AI must not fabricate prices, override calculations, hide uncertainty, or present estimates as guarantees
Summarize trade-offs	—	Summarize visible warnings and assumptions in accessible language	Understand and weigh trade-offs	All numbers in explanation must exactly match engine output
Show weather context	Display weather data with source, timestamp, coverage label	May phrase weather impact in accessible language	Interpret weather relevance	Weather must not trigger automated recommendations
Handle connectivity loss	Cache management; offline banner; graceful degradation	None	Decide whether to trust cached data	Cached data must be labeled with timestamp; never presented as live
Make selling decision	—	—	**Farmer alone decides**	System must not display "recommended," "best," or "should sell" anywhere

Boundary rule: The AI layer receives only validated deterministic output as input. There is no code path by which the AI can modify a calculated value, invent a quote, or suppress a warning.

I. Data and Integration Design

I.1 Data Classification

Category	Description	Examples	Status
1. Confirmed existing functionality	Artifacts present in this repository and verified	6 JSON test fixtures; 15 Markdown documents; calculation formula specified	Exists
2. Proposed design	Specified but not implemented	Deterministic calculation engine; AI explanation layer; offline cache; comparison UI	Proposed
3. Simulated development/test behaviour	Data created to exercise test scenarios	All fixture market data (prices ₹14–24/kg, costs, timestamps)	Simulated — labeled
4. Future integration requiring verification	External systems not yet connected	eNAM API; IMD/OpenWeatherMap API; LLM service	Future — not connected

I.2 Simulated Fixture Data

Six fixture files exist in `tests/fixtures/` and are version-controlled:

Fixture	Purpose	Key values
`at01-basic-calculation.json`	AT-01: basic calculation	₹24/kg, costs ₹500 total, 100 kg → ₹19/kg, ₹1,900
`at02-tie-on-net-proceeds.json`	AT-02: transport erases price advantage	A: ₹22/kg ₹200 costs → ₹20/kg; B: ₹24/kg ₹500 costs → ₹19/kg

Fixture	Purpose	Key values
`at03-missing-quote.json`	AT-03: missing quote	Market B: `quoted_price_per_kg: null`
`at04-stale-quote.json`	AT-04: stale quote	Timestamp 6 hours before evaluation time; threshold 4h
`at05-offline-cached.json`	AT-05: offline with cache	`quote_source: "cached"`, `connectivity: "offline"`
`at06-weather-missing.json`	AT-06: weather unavailable	`weather: null`

All fixtures are labeled `"simulated"` or contextually marked. None represent verified real-world data.

I.3 Market-Price Data Source Strategy

Tier	Source	Role	Verification status
1 (V1 primary)	Simulated fixtures	Development, testing, demonstration	**Exists** — in repository
2 (prototype)	eNAM / state mandi API	Live quote demonstration	**Not verified** — access, coverage, freshness, rate limits unknown
3 (weather)	IMD / OpenWeatherMap	Contextual forecast	**Not connected** — structure defined only
4 (fallback)	LocalStorage cache, max age proposed 4 hours	Offline / intermittent operation	**Proposed** — behavior specified in AT-05, not implemented

No external API is claimed to be operationally connected.

I.4 Freshness Policy

Parameter	Value	Classification
Default maximum quote age	4 hours	**Proposed policy** — configurable; not a universal agricultural standard
Behavior when exceeded	`data_freshness: "stale"`; age displayed; warning shown	Defined (FR-05, AT-04)
Behavior when timestamp missing	`data_freshness: "unknown"`; warning shown	Defined (FR-05)

I.5 Source and Timestamp Preservation

Every quote must carry: `quote_source` (string), `quote_timestamp` (ISO 8601), and a method indicator (live / cached / manual / simulated). These are preserved from input through to display. No transformation discards provenance metadata.

I.6 Offline / Cache Strategy

- Cache stored client-side (LocalStorage or Service Worker) — **proposed implementation**.
- On offline detection: load cache if age < threshold; display offline banner with cache timestamp.
- If cache stale or empty: allow manual quote entry; display "All data manually entered" warning.
- Core calculations remain available on cached or manual data.

J. Constraints and Assumptions

J.1 Confirmed Case Evidence (Simulated Competition Figures)

Preserved from `docs/02-evidence-ledger.md`. All are **simulated case evidence**, not verified real-world statistics.

ID	Claim	Limitation
E01	Mandi quotes range ₹14–24/kg across 4 markets (same morning)	Obtainability by a specific farmer uncertain
E02	90 km market transport/loading removes ₹4–6/kg of price advantage	Cost variation by volume and shared transport unknown
E03	2 of 5 villages report unreliable mobile data	Best offline strategy (SMS/IVR/WhatsApp/offline/intermediary) unknown
E04	Farmers trust local traders/neighbours more than unfamiliar apps	What evidence would change trusted routines unknown
E05	Heavy rain possible within 36 hours	Local vs district forecast divergence unknown
E06	8–15% post-harvest loss in oversupply weeks	Avoidable portion unknown

J.2 Project Assumptions

ID	Assumption	Validation needed
A01	Farmers can estimate or obtain transport/loading costs for their route and volume	Survey / field validation
A02	Quotes have identifiable source and timestamp (even if simulated)	Prototype integration must preserve provenance
A03	Net proceeds comparison is a meaningful decision criterion for target farmers	Usability testing with proxy users
A04	English-only UI is acceptable for V1	Stakeholder confirmation
A05	Simulated/test dataset is sufficient for Round 1 demonstration	If live API available, integrate as labeled prototype layer

J.3 Technical and Operational Constraints

Constraint	Implication
Limited connectivity (E03)	System must function degraded: cached data + warnings; manual entry; no claim of complete offline operation
Missing or stale quotes (E01, U01, U03)	Markets with missing data excluded from ranking; stale data flagged; 4-hour threshold is proposed, configurable
Variable transport and handling costs (E02, U02)	Costs are user-entered estimates; never silently defaulted to zero; sensitivity note in explanation
Incomplete source coverage	V1: 1 crop, ≤5 mandis; no claim of comprehensive coverage
Limited initial crop/market coverage	Single-crop scope is deliberate (V1 boundary), not a limitation of the architecture
Dependence on external AI service	Explanation layer has template fallback; core comparison works without AI
Dependence on external weather service	Weather is optional context; system functions without it (AT-06)
Unverified live integrations	No live API claimed as connected; simulated fixtures are primary data source for V1

J.4 Unresolved Questions

ID	Question	V1 treatment
U01	Are quoted prices actually obtainable by the farmer?	Label as "quoted/reference"; farmer verification prompt
U02	How do transport costs vary by quantity, distance, shared transport?	User-entered estimate; sensitivity note
U03	How fresh must a quote be?	Proposed threshold 4 hours; configurable; explicit warning
U04	What quality loss from delaying sale?	Excluded from V1; future research
U05	How reliable is district forecast locally?	Coverage label; no decision automation
U06	Which farmers need SMS/voice/IVR?	V1: cached data + warnings; accessibility modes = future

K. Failure Handling and Fallbacks

Failure ID	Trigger	System response	User-facing message / warning	Calculations continue?	Data-integrity safeguard	Recovery action
FH-01	Missing price quote (`quoted_price_per_kg` is null)	Mark `data_freshness: "missing"`; exclude from ranking	"No quote available for [market]. Cannot compare." Display "—" for price fields.	Yes — for other markets with sufficient data	Never invent a price; market shown with explicit gap	User may enter quote manually if available from another source
FH-02	Conflicting quotes from multiple sources	Show all values with their sources; flag conflict; do not auto-select	"Two sources report different prices for [market]. Review both."	Yes — calculation runs but market flagged	Conflict surfaced; no silent resolution; user decides which to trust	User selects preferred source or verifies with trader
FH-03	Stale quote (age > freshness threshold)	Mark `data_freshness: "stale"`; display age and timestamp	"Quote is N hours old (threshold: X hours). Verify before relying on this comparison."	Yes — with stale warning attached	Stale status always visible; never hidden	User verifies current price with market/trader
FH-04	Unknown transport or handling cost	Mark market "costs unknown"; exclude from net-proceeds ranking	"Selling costs unknown for [market]. Cannot calculate net proceeds."	Yes — for markets with known costs	**Costs never silently defaulted to zero**	User enters estimated cost, or market left unranked
FH-05	Quantity ≤ 0 or non-numeric	Reject input before calculation	"Quantity must be greater than zero."	No — calculation not started	Validation before any arithmetic	User corrects input
FH-06	Weather service unavailable or returns null	Display unavailable message with last-attempt timestamp	"Weather data unavailable. Showing market comparison without weather context."	Yes — weather is optional context	Weather never fabricated; absence disclosed	Retry when connectivity restored; comparison unaffected

Failure ID	Trigger	System response	User-facing message / warning	Calculations continue?	Data-integrity safeguard	Recovery action
FH-07	Network failure (no connectivity, no cache)	Allow manual data entry; display offline status	"Offline — no cached data. Enter quotes manually."	Yes — on manual inputs	Manual inputs labeled "user-entered"; not presented as sourced data	Reconnect when possible; cache refresh on next successful fetch
FH-08	Network failure with valid cache	Load from cache; display offline banner	"Offline — showing cached data from [timestamp]."	Yes — on cached inputs	Cache timestamp always visible; cached data never labeled "live"	Refresh cache on reconnection
FH-09	AI explanation service unavailable	Fallback to template explanation; deterministic table still shown	"Explanation service unavailable — review the numbers directly."	Yes — deterministic calculation unaffected	Fallback template uses only engine output numbers; no AI-generated content	Retry AI service; comparison remains valid regardless
FH-10	Invalid or incomplete source data (malformed JSON, wrong units)	Reject source; flag data-quality error; prompt alternative input	"Data source returned invalid information. Please enter values manually."	Yes — with manual inputs	Invalid data never partially processed; rejected cleanly	User manual entry or source correction

---

L. Acceptance Criteria and Test Plan

L.1 Test Fixtures

All six fixtures exist and are verified in `tests/fixtures/`:

Fixture file	Verified	Key test values
`at01-basic-calculation.json`	✅ Exists	₹24/kg, ₹500 total costs, 100 kg
`at02-tie-on-net-proceeds.json`	✅ Exists	A: ₹22/kg ₹200; B: ₹24/kg ₹500; 100 kg
`at03-missing-quote.json`	✅ Exists	Market B quote = null
`at04-stale-quote.json`	✅ Exists	2026-10-14T02:00:00Z (6h old vs 4h threshold)
`at05-offline-cached.json`	✅ Exists	source="cached", connectivity="offline"
`at06-weather-missing.json`	✅ Exists	weather = null

L.2 Acceptance Test Definitions

Test ID	Requirements covered	Preconditions	Input fixture	Steps	Expected result	Pass/fail criterion	Status
AT-01	FR-03, FR-04, NFR-01	Calculation engine available; valid inputs	`at01-basic-calculation.json`: ₹24/kg, transport ₹300, loading ₹100, unloading ₹100, qty 100 kg	1. Load fixture 2. Run calculation engine 3. Record output	selling_costs_per_kg = ₹5.00; net_realization_per_kg = ₹19.00; total_net_proceeds = ₹1,900	Engine returns exactly `net_realization_per_kg: 19.00` and `total_net_proceeds: 1900.00`	**Not run**
AT-02	FR-03, FR-04, NFR-01	Two markets with valid data	`at02-tie-on-net-proceeds.json`: A ₹22/kg ₹200 costs; B ₹24/kg ₹500 costs; qty 100 kg	1. Load fixture 2. Calculate both markets 3. Rank by net proceeds	Mandi A: ₹20.00/kg net, ₹2,000 total; Mandi B: ₹19.00/kg net, ₹1,900 total; Mandi A ranked first	Mandi A ranked above Mandi B; no output claims Mandi B is better; displayed net values match ₹20 and ₹19 exactly	**Not run**
AT-03	FR-02, FR-05, NFR-04	One market has missing quote	`at03-missing-quote.json`: Market B `quoted_price_per_kg: null`	1. Load fixture 2. Run data-quality check 3. Observe comparison display	Market B: `data_freshness: "missing"`; excluded from ranking; "—" displayed; no price invented	No fabricated price for Market B; Market B not ranked; warning visible	**Not run**
AT-04	FR-02, FR-05, NFR-04	Quote age exceeds threshold	`at04-stale-quote.json`: timestamp 2026-10-14T02:00:00Z; threshold 4h; evaluation time 2026-10-14T08:00:00Z	1. Load fixture 2. Run freshness check 3. Observe warning display	`data_freshness: "stale"`; warning "Quote is 6 hours old (threshold: 4h)" visible; timestamp displayed	Stale warning visible; timestamp shown; stale status in output	**Not run**
AT-05	FR-09, NFR-04	Network offline; cached quote available	`at05-offline-cached.json`: source="cached", connectivity="offline"	1. Simulate offline 2. Load fixture 3. Observe banner and data presentation	Banner "Offline — showing cached data from 2026-10-14T06:00:00Z"; calculation functions; cached data not labeled "live"	Banner shown with timestamp; calculations produce correct results; no "live" label on cached data	**Not run**

Test ID	Requirements covered	Preconditions	Input fixture	Steps	Expected result	Pass/fail criterion	Status
AT-06	FR-06, FR-07, NFR-04	Weather data unavailable	`at06-weather-missing.json`: `weather: null`	1. Load fixture 2. Run system 3. Observe weather and explanation	"Weather data unavailable" displayed; no weather-dependent recommendation made; market comparison proceeds normally	No weather recommendation; limitation disclosed; comparison complete with available data	**Not run**

L.3 Test Execution Status

All six tests are marked "Not run." Round 1 requires test definitions with explicit pass/fail criteria. Execution occurs in Build Round once the deterministic engine and UI are implemented. No test results are claimed.

L.4 Evaluation Targets (Proposed, Not Measured)

Dimension	Proposed target	Classification
Calculation correctness	100% match against manual verification	Target — not measured
Missing/stale data handling	All flags visible; zero fabricated values	Target — not measured
Explanation ↔ calculation consistency	Zero numerical discrepancies	Target — not measured
Understandability (proxy users)	>80% identify higher-net market and key assumption	Target — not measured
Performance	Comparison renders within 2 seconds (5 markets, mid-range device)	Target — not measured
Accessibility	Functional at 320px viewport; 2G simulation	Target — not measured

No evaluation result is reported as achieved. These are proposed targets for Build Round validation.

M. Version 1 Scope and Exclusions

Included in V1	Excluded from V1	Reason for exclusion	Possible future version
Net-proceeds comparison for 1 crop, up to 5 mandis	Multi-crop, nationwide mandi coverage	Scope control for feasible, testable V1	V2: multi-crop with verified data coverage
Deterministic calculation engine	ML-based price prediction	Prediction requires training data and validation not available	V3+ (requires research)
Simulated/test dataset with provenance	Guaranteed live API dependency	Live API access not verified; reproducible tests prioritized	V2: verified live API integration
Source/timestamp/freshness on every quote	Real-time streaming quotes	Streaming requires backend infrastructure; V1 is client-side	V2+ (requires backend)
Stale/missing/conflict data flags	Automatic resolution of conflicts	Auto-resolution risks hiding data-quality problems; transparency required	V2: source-priority rules with user confirmation
Weather context (source, time, coverage)	Harvest-timing recommendation	U04: quality loss data unknown; would require crop-quality models	V3 (requires field research)

Included in V1	Excluded from V1	Reason for exclusion	Possible future version
Offline cached data with warnings	SMS / IVR / WhatsApp / voice interfaces	U06: accessibility needs unverified; channels require partner integrations	V2+: accessibility modes based on field validation
English UI	Hindi / local language interface	A04: V1 scope decision; translations require verified numerical round-trip	V2: Hindi with round-trip numerical verification
Farmer decision authority explicit	Autonomous selling or harvest decisions	Case boundary: system must not replace authorised advice	Not planned — permanent boundary
Transparent assumptions and limitations	Guaranteed outcomes (price, income, waste reduction)	Case boundary: no unsupported guarantees	Not planned — permanent boundary

N. Version 1 Implementation Plan

Based on the existing 15-working-day proposal from `specs/01-solution-srs.md`.

Phase	Days	Tasks	Deliverables	Dependencies	Validation / testing	Completion criteria	Key risks
1. Scope confirmation	Day 1	Freeze problem statement, evidence ledger, V1 boundary; confirm FR/NFR IDs	Signed-off scope document	Round 1 submission complete	Review against SRS §E, §F	All requirements IDs confirmed; no scope ambiguity	Scope creep during build
2. Data & calculation core	Days 2–3	Implement deterministic engine (`calculate_net_proceeds`); define data schemas; load fixture dataset	Tested calculation engine; schema definitions; fixtures wired	Phase 1	Unit tests against AT-01, AT-02 expected values; manual recalculation	AT-01, AT-02 calculation assertions pass in unit tests	Off-by-one or unit errors in formula implementation
3. Retrieval & freshness	Days 4–5	Build quote loader (fixture + optional API adapter); implement freshness checker; implement cache layer	Quote fetcher; freshness evaluation; LocalStorage cache	Phase 2	AT-04 (stale), AT-03 (missing) unit tests; conflict detection tests	Freshness threshold correctly flags stale quotes; missing quotes correctly excluded	API access delays if live integration attempted
4. Comparison UI	Days 6–8	Build React input form; comparison table; warning display; weather panel; responsive layout	Mobile React app with FR-01, FR-02, FR-04, FR-07, FR-08	Phase 2, 3	Input validation tests; visual inspection; AT-03 UI display check	All Must-have FRs demonstrated in UI; responsive at 320px	UI complexity; responsive layout issues on low-end devices

Phase	Days	Tasks	Deliverables	Dependencies	Validation / testing	Completion criteria	Key risks
5. Explanation layer	Days 9–10	Build AI explanation prompt; implement template fallback; consistency checker	Explanation generator with fallback	Phase 4	AT-06 consistency test: explanation numbers == engine numbers	Zero numerical discrepancies between explanation and engine output	LLM hallucination; fallback template quality
6. Resilience & offline	Days 11–12	Implement Service Worker / cache; offline banner; graceful degradation for API failures	Offline-capable behavior; AT-05 support	Phase 3, 4	AT-05 offline test; network-failure simulation	Offline banner with correct timestamp; calculations function on cached data	Service Worker complexity; cache invalidation
7. Acceptance testing	Day 13	Execute AT-01 through AT-06; record results; fix critical defects	Test execution records; defect log	Phases 2–6 complete	Run all 6 tests against fixtures	All Must-have FR tests pass; defects documented	Defects discovered late; time pressure
8. Evaluation & documentation	Day 14	Run evaluation plan dimensions; document results; update AI interaction log	Evaluation results; updated documentation	Phase 7	Proxy-user understandability test; consistency review	Evaluation targets assessed; results honestly recorded	Proxy-user availability; time for evaluation
9. Demonstration preparation	Day 15	Prepare demo walkthrough; export SRS/PDF; final consistency check	Demo script; exported documents; final review	Phase 8	Full consistency audit per §7 of this document	All artifacts consistent; demo script tested	Export formatting issues; last-minute inconsistencies

Parallelization opportunities: Phases 4 (UI) and 5 (explanation layer) can partially overlap after Phase 2 completes. Phase 3 can start in parallel with Phase 4 if schema is defined.

Status: All phases are **planned, not completed.** No phase reports completion.

O. Technical Feasibility and Round 2 Build Plan

O.1 Technology Choices

All choices below are **proposals** informed by existing project documentation unless noted.

Component	Proposed technology	Rationale	Classification
Frontend	React (mobile-first)	Team expertise documented in project files; component ecosystem; works on low-end devices	Proposal (consistent with project docs)
Calculation engine	Python or JavaScript pure functions	Zero dependencies; independently unit-testable; no LLM involvement	Proposal (consistent with project docs)

Component	Proposed technology	Rationale	Classification
AI explanation	Template-based + LLM API (e.g., GPT-4o-mini class)	Templates ensure correctness; LLM adds language flexibility	Proposal (consistent with project docs)
Client-side storage	LocalStorage or Service Worker cache	No server dependency; offline-capable; privacy-preserving	Proposal (consistent with project docs)
Hosting	Static hosting (Vercel/Netlify class)	No backend required for V1; free tier sufficient	Proposal — not established in source docs

O.2 Proposed Module Breakdown

```
krishived-market-decision-support/
├── src/
│   ├── engine/              # Deterministic calculation (pure functions)
│   │   ├── calculate.py     # calculate_net_proceeds()
│   │   ├── freshness.py     # evaluate_freshness()
│   │   └── ranking.py       # rank_by_net_proceeds()
│   ├── data/                # Data layer
│   │   ├── loader.py        # Fixture/API quote loader
│   │   ├── validator.py     # Input validation, conflict detection
│   │   └── cache.py         # LocalStorage cache management
│   ├── explain/             # AI explanation layer
│   │   ├── template.py      # Fallback template generator
│   │   └── llm.py           # LLM prompt + consistency check
│   ├── ui/                  # React components
│   │   ├── InputForm.jsx    # FR-01 data entry
│   │   ├── ComparisonTable.jsx  # FR-02, FR-04 display
│   │   ├── WarningPanel.jsx # FR-05 warnings
│   │   ├── WeatherPanel.jsx # FR-07 weather context
│   │   └── OfflineBanner.jsx # FR-09 connectivity status
│   └── App.jsx
├── tests/
│   ├── fixtures/            # 6 JSON test fixtures (exist)
│   └── test_engine.py       # AT-01 to AT-06 test implementations
├── docs/
│   ├── SUBMISSION.md        # Round 1 submission (exists)
│   └── specs/               # SRS documents (exist)
└── package.json
```

Repository status: This structure is a **proposal**. The current repository contains documentation and test fixtures only; no source code has been written.

O.3 Minimum Viable Demonstrable Prototype

A demonstrable V1 prototype requires:

Component	Minimum requirement	Why it's minimum
Input form	Crop, quantity, 2–5 mandis with price and cost fields	Core user journey entry point
Deterministic engine	`calculate_net_proceeds()` working and unit-tested	Core value; all comparisons depend on it
Comparison table	Ranked display with freshness flags and warnings	Demonstrates transport-adjusted insight (AT-02)
AI explanation	Template fallback at minimum; LLM optional	FR-06 fulfilled by template; LLM is enhancement
Offline banner	Cached-data timestamp display	AT-05 demonstrates connectivity handling

Component	Minimum requirement	Why it's minimum
6 test fixtures	Already exist	Test infrastructure ready

Estimated prototype scope: Input form + calculation engine + comparison table + template explanation + offline banner = demonstrable in the 15-day plan.

O.4 Core Dependencies

Dependency	Purpose	If unavailable
React (npm)	UI framework	Alternative: vanilla JS (proposal)
Python or JS runtime	Calculation engine	Already available
LLM API (optional)	AI explanation enhancement	Template fallback is sufficient for V1
Browser LocalStorage	Cache	Alternative: in-memory cache (reduced persistence)

O.5 Development Milestones

Milestone	Target day	Evidence required
M1: Scope frozen	Day 1	Confirmed requirement IDs in this document
M2: Engine unit tests pass	Day 3	Test output showing AT-01, AT-02 assertions passing
M3: UI renders comparison	Day 8	Screenshot of comparison table with fixture data
M4: Explanation consistency verified	Day 10	Automated check: explanation numbers == engine numbers
M5: AT-01 to AT-06 executed	Day 13	Test execution log with pass/fail results
M6: Evaluation assessment complete	Day 14	Evaluation results document with honestly reported outcomes

O.6 Testing Strategy

1. **Unit tests**: Calculation engine tested against fixture expected values (AT-01, AT-02).
2. **Integration tests**: Data-quality checks tested against missing/stale/conflict fixtures (AT-03, AT-04, AT-06).
3. **Offline tests**: Cache behavior tested with network simulation (AT-05).
4. **Consistency tests**: Explanation output compared against engine output for numerical accuracy.
5. **Manual evaluation**: Proxy-user walkthrough for understandability (proposed target: >80%).

O.7 Evidence Required to Prove Implementation Works

Claim	Required evidence
"Calculation engine works"	Unit test output showing AT-01/AT-02 pass with exact expected values
"Data-quality flags work"	AT-03/AT-04 test output showing correct freshness status and warnings
"Offline mode works"	AT-05 test output showing banner and correct cached calculations
"Explanation is consistent"	Automated comparison showing zero numerical discrepancies
"UI is usable"	Screenshot evidence + proxy-user test results
"All tests pass"	Complete test execution log with date, result, defects

O.8 Features Deferred Until Core Workflow Is Validated

- Hindi interface (pending A04 validation and round-trip numerical verification)
- Live mandi API integration (pending access verification)
- Weather API integration (pending source verification)
- SMS/IVR/WhatsApp channels (pending U06 resolution)
- Harvest-timing comparison (pending U04 research)
- Multi-crop support (pending scope expansion decision)
- Performance optimization (pending NFR-09 measurement)

P. Risks and Mitigations

Risk ID	Description	Likelihood	Impact	Mitigation	Contingency	Residual risk
R-01	Unreliable or inaccurate market data (quotes not obtainable, stale at source)	Medium	High	Label all quotes with source and freshness; farmer verification prompt; "quoted/reference" labeling	Exclude unreliable quotes from ranking; show "cannot compare"	Farmer may still act on stale info despite warnings
R-02	Stale quotes mislead farmer into wrong decision	Medium	High	4-hour proposed threshold; visible age display; prominent stale warnings	Farmer verifies with trader before selling	Threshold may be inappropriate for some crops/markets
R-03	User-entered transport costs inaccurate	High	Medium	Clear labeling as estimate; sensitivity note in explanation; cost input is explicit	Show both quoted price and net proceeds so farmer sees cost impact	Net proceeds may be wrong if estimates are significantly off
R-04	Connectivity limitations prevent data retrieval (E03)	High	Medium	Offline cache with timestamp; manual entry fallback; offline banner	Calculations continue on cached/manual data	Cache may be stale; manual entry may introduce errors
R-05	AI explanation hallucinates or misrepresents numbers	Medium	High	AI receives only validated JSON; consistency check (explanation vs engine); template fallback	Disable AI explanation; show deterministic table only	Subtle phrasing errors may survive numerical check
R-06	Misleading comparison (farmer misinterprets ranking)	Medium	High	No "recommended" or "best" labels; assumptions panel visible; farmer verification prompt	Disclaimers on every output; decision-authority statement	Farmer may still over-trust the tool
R-07	Integration delays (live API access not granted)	Medium	Low	Simulated fixtures as primary source; no dependency on live API for V1	Demonstrate with fixtures; document API as "future integration"	Prototype may appear less impressive without live data

Risk ID	Description	Likelihood	Impact	Mitigation	Contingency	Residual risk
R-08	Limited testing time before Round 2 deadline	Medium	Medium	15-day plan with phase gates; Must-have FRs prioritized	Ship with Must-have FRs tested; Should-have documented but untested	Incomplete test coverage for Should-have features
R-09	Bias in data or explanations (regional, crop, or market-size)	Low	Medium	Simulated fixture data deliberately neutral; explanation prompts prohibit comparative value language; ranking is arithmetic	Audit any future live data for representational bias	Undetected bias may persist in future live-data versions
R-10	Incorrect recommendation consequence (farmer loses income)	Low	High	System never makes recommendations; farmer retains authority; verification prompt before exit; disclaimers on weather/financial screens	Educational messaging about tool limitations	Farmer may still attribute bad outcomes to the tool

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Q. Deliverables and Definition of Done

Q.1 Round 1 Submission Deliverables (Current Stage)

Deliverable	Definition of done	Status
Basic SRS / Implementation Plan (this document)	All sections A–Q present; consistent with `SUBMISSION.md` ; quality audit completed; no unsupported claims	**Created** — this document
PDF export of this document	PDF renders all sections, tables, and a non-interactive workflow diagram (Mermaid rendered to SVG, not source text)	**Created** — `specs/Basic_SRS_Implementation_Plan.pdf` (32 pages; diagram spans pages 13–14 due to diagram height)
`SUBMISSION.md` (master Round 1 document)	14 sections complete; 15-item checklist verified; 91-item audit passed (non-blocked items)	**Created** — verified
Evidence ledger with E/A/U identifiers	Case evidence, assumptions, unknowns separated and labeled	**Created** — `docs/02-evidence-ledger.md`
Problem-framing comparison	≥3 framings compared; selection justified with evidence	**Created** — `docs/01-problem-framing.md`
Acceptance test definitions (AT-01 to AT-06)	Scenarios, inputs, expected results, pass/fail criteria documented; fixtures exist	**Created** — `tests/acceptance-tests.md` + 6 fixtures
AI interaction log and decision index	8 interactions documented; 8 decisions traced; honest session references	**Created** — `ai-trail/`
Existing-component disclosure	All components declared accurately; prior work acknowledged	**Created** — `ai-trail/existing-components.md`

Q.2 Round 2 Build Deliverables (Future Stage)

Deliverable	Definition of done	Status
Functional prototype	Input form + calculation engine + comparison table + explanation + offline banner demonstrated	**Not started**
Executed acceptance tests	AT-01 to AT-06 run with recorded results and defect log	**Not started** — tests marked "Not run"
Evaluation results	Evaluation plan dimensions assessed with honestly reported outcomes	**Not started**
Test evidence (screenshots, logs)	Visual and log evidence that implementation works	**Not started**
GitHub repository with code	Source code committed; build succeeds; tests pass	**Not started** — documentation-only repo currently
Round 2 demo material	Demo script, walkthrough, exported documents	**Not started**

Q.3 Claims Requiring Evidence (Not Yet Supported)

Claim	Required evidence	Current status
"Calculation engine produces correct results"	Unit test output	Not yet available
"Acceptance tests pass"	Test execution log	Tests marked "Not run"
"System works offline"	AT-05 execution evidence	Not yet available
"Explanation matches calculations"	Consistency check output	Not yet available
"Farmer can understand the interface"	Proxy-user test results	Not yet available
"Live API integration works"	API access confirmation + integration test	Not confirmed — labeled "future integration"

Quality Audit

Checklist per Section 7 of the master prompt:

#	Check	Result
1	All 12 competition-required topics covered (§2 of master prompt: users, inputs, outputs, workflow, FRs, AI resp., human resp., constraints, failure handling, acceptance criteria, V1 scope, implementation plan)	✅ Sections B, D, G, E, H, J, K, L, M, N — all present
2	Every functional requirement has a unique ID	✅ FR-01 to FR-10
3	Every non-functional requirement has a unique ID	✅ NFR-01 to NFR-09
4	Every acceptance test has explicit expected results	✅ AT-01 to AT-06 all have expected results and pass/fail criteria
5	AT-02 calculation correct: A = ₹20/kg, B = ₹19/kg	✅ Verified in §G.3: A: 22−2=20; B: 24−5=19
6	Missing costs not silently treated as zero	✅ FH-04: "Costs never silently defaulted to zero"; FR-04 excludes "costs unknown" markets

#	Check	Result
7	Stale, simulated, cached, conflicting data clearly distinguished	✅ Freshness enum (`current` / `stale` / `missing` / `unknown`); "simulated" labels throughout; cache source identified; FH-02 conflict handling
8	AI explanations cannot override validated calculations	✅ §H boundary rule: "no code path by which AI can modify a calculated value"
9	Farmer remains final decision-maker	✅ §H final row; §G.1 step 11; case boundary respected
10	4-hour freshness policy consistent throughout	✅ Used in FR-05, FH-03, I.4, fixtures, tests — consistently proposed, not confirmed as standard
11	Planned work not represented as completed	✅ All implementation phases marked "planned, not completed"; all tests "Not run"; §Q.2 "Not started"
12	All V1 exclusions match existing project scope	✅ §M matches SUBMISSION.md §10 and docs/03 scope tables
13	Implementation plan feasible within 15-working-day schedule	✅ 9 phases mapped to 15 days with dependencies identified
14	Missing fixtures, source files, test logs, integrations disclosed	✅ §I.1 data classification; §I.3 "not verified" API status; §Q.3 claims requiring evidence
15	No unsupported claims or invented results remain	✅ No team names, no API integrations claimed, no test results claimed, no field trials, no performance measurements claimed

Audit result: 15/15 checks passed. No unresolved consistency issues identified.

Submission Readiness Checklist

Requirements documented

- [x] Users and stakeholders (§B)
- [x] System inputs with validation rules (§D.1)
- [x] System outputs with warnings (§D.2)
- [x] End-to-end workflow with failure flows and diagram (§G)
- [x] Functional requirements FR-01 to FR-10 (§E)
- [x] Non-functional requirements NFR-01 to NFR-09 (§F)
- [x] AI responsibilities matrix (§H)
- [x] Human responsibilities matrix (§H)
- [x] Technical and operational constraints (§J.3)
- [x] Failure handling FH-01 to FH-10 (§K)
- [x] Acceptance criteria AT-01 to AT-06 (§L)
- [x] V1 scope and exclusions (§M)
- [x] Implementation plan for Round 2 (§N, §O)

Artifacts that must be created (not yet existing)

- [ ] Deterministic calculation engine source code
- [ ] React UI source code

- [ ] AI explanation prompt/template implementation
- [ ] Offline cache implementation
- [ ] Test execution scripts
- [ ] GitHub repository with committed source code

Tests that must actually be run

- [ ] AT-01 (calculation correctness) — currently "Not run"
- [ ] AT-02 (transport-erases-advantage) — currently "Not run"
- [ ] AT-03 (missing quote handling) — currently "Not run"
- [ ] AT-04 (stale quote handling) — currently "Not run"
- [ ] AT-05 (offline with cache) — currently "Not run"
- [ ] AT-06 (weather unavailable) — currently "Not run"

Claims that require evidence

- [ ] "Engine produces correct results" → needs unit test output
- [ ] "All acceptance tests pass" → needs test execution log
- [ ] "System handles offline gracefully" → needs AT-05 execution evidence
- [ ] "Explanation is consistent with calculations" → needs consistency check output
- [ ] "Interface is understandable" → needs proxy-user test results
- [ ] "Live API integration available" → needs access confirmation (currently not verified)

Items that remain unresolved

- [ ] Live mandi API access (U-dependent; not verified)
- [ ] Weather API access (not verified)
- [ ] Hindi interface scope decision (A04 pending stakeholder confirmation)
- [ ] Optimal freshness threshold for different crops/markets (U03)
- [ ] Best offline access strategy: SMS/IVR/WhatsApp/offline/intermediary (U06)
- [x] Portal packaging artifacts: this document exported to `specs/Basic_SRS_Implementation_Plan.pdf` with rendered workflow diagram
- [ ] Portal-specific submission format requirements (portal naming/size rules — manual verification on Unstop portal)

This document is a specification and plan, not a statement of implementation readiness. The supporting source code, executed tests, and verified integrations do not yet exist.

Document prepared for AI Synergy Hackathon 2026 Round 1. All simulated competition figures are labeled as such. All proposed targets are labeled as proposed. All test statuses reflect actual state: "Not run."